文献汇报

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Deep Dual-Channel Neural Network for Image-Based Smoke Detection

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IEEE Transactions on Multimedia

CiteScore

CiteScore	SJR	SNIP	CiteScore排名			
			学科	分区	排名	百分位
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			大类：Computer Science 小类：Signal Processing	Q1	5 / 99	95%
			大类：Computer Science 小类：Computer Science Applications	Q1	22 / 569	96%
			大类：Engineering 小类：Electrical and Electronic Engineering	Q1	28 / 661	95%

DCNN网络结构

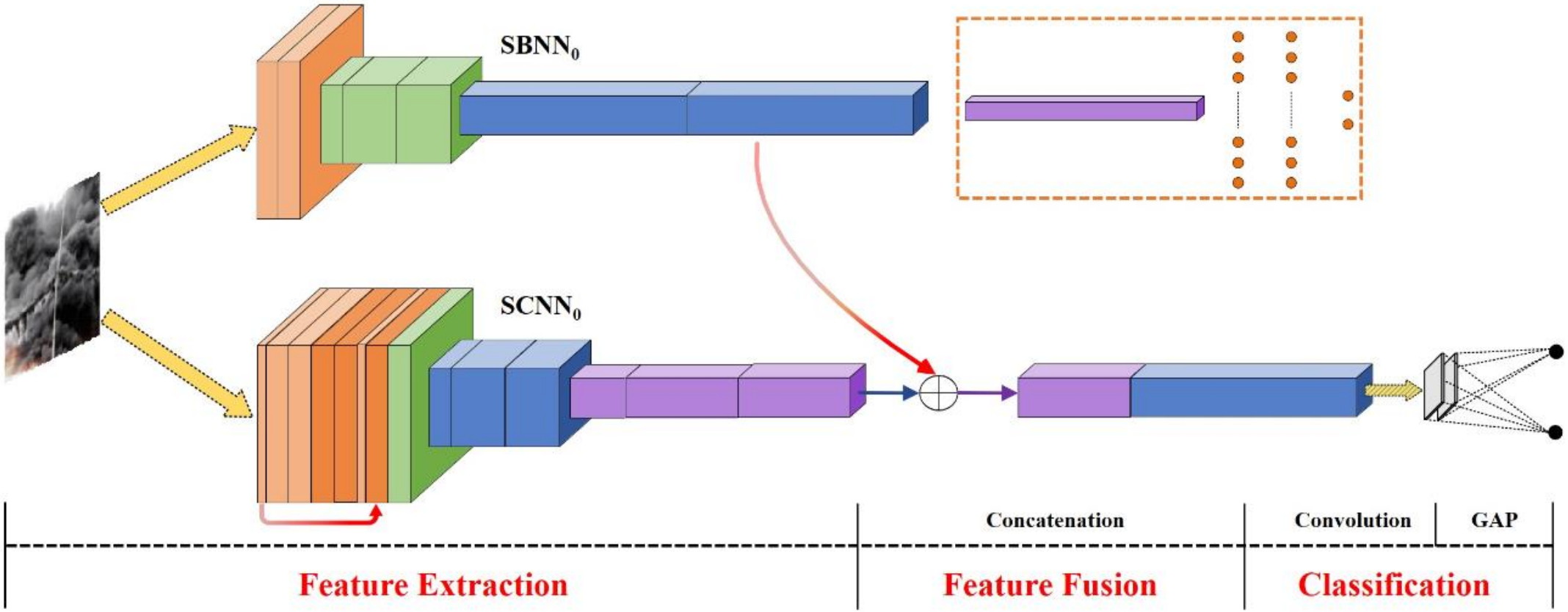


Fig. 3: The basic architecture of DCNN, including the dual channels of deep subnetworks for feature extraction, a concatenation operation for feature fusion, as well as a convolution layer and a global pooling layer for classification. 'GAP' stands for the global average pooling operation. For the sake of clarity, we remove the operation symbols from the original subnetworks and only keep the feature blocks. The top channel is the SBNN$_0$ and the bottom channel is the SCNN$_0$.

DCNN 可视化

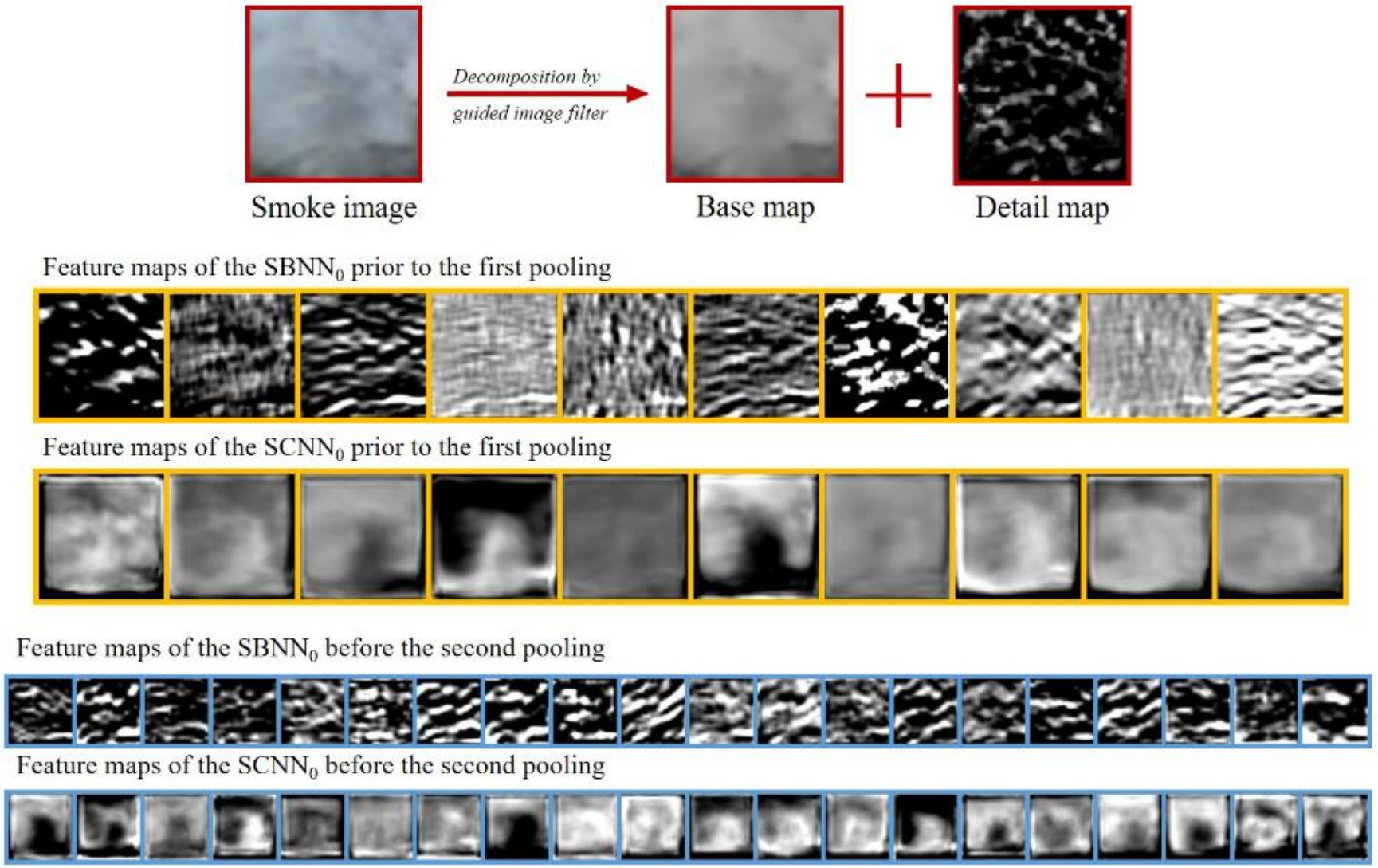


Fig. 7: Illustration of typical intermediate feature maps extracted using SBNN$_0$ and SCNN$_0$ prior to two pooling layers.

DCNN 劣势

Fig. 8: Falsely detected samples from Set-1 and Set-2.

对纹理信息要求大

DCNN DNCNN结果

$$\mathrm{DR} = \frac{P_p}{Q_p} \times 100\%$$

$$\mathrm{FAR} = \frac{N_p}{Q_n} \times 100\%$$

$$\mathrm{AR} = \frac{P_P + N_n}{Q_n + Q_p} \times 100\%$$

TABLE VII: Comparison of DCNN with its two components and DNCNN.

Methods		DNCNN [22]	SBNN (Prop.)	SCNN (Prop.)	DCNN (Prop.)
Set1	AR	97.8%	98.3%	98.6%	99.7%
	DR	95.2%	97.3%	97.6%	99.5%
	FAR	0.48%	0.96%	0.84%	0.12%
Set2	AR	98.0%	98.7%	98.5%	99.4%
	DR	96.3%	98.4%	97.2%	99.0%
	FAR	0.48%	0.98%	0.48%	0.24%

A Deep Normalization and Convolutional Neural Network for Image Smoke Detection

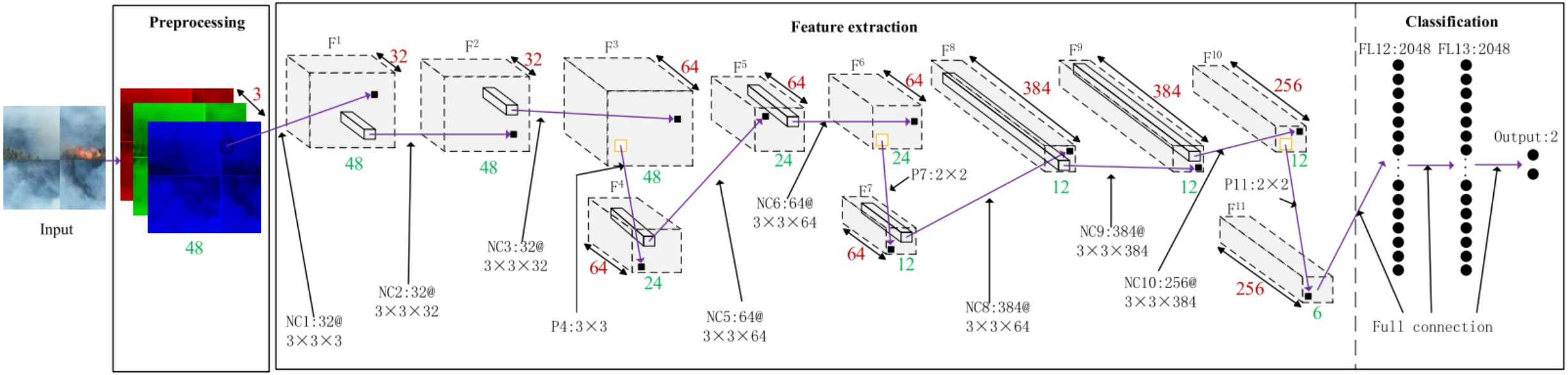
ZHIJIAN YIN[1], BOYANG WAN[1], FEINIU YUAN[2], (Senior Member, IEEE), XUE XIA[2], AND JINTING SHI[2,3]

IEEE access

CiteScore

CiteScore	SJR	SNIP	CiteScore排名			
			学科	分区	排名	百分位
4.96	0.609	1.718	大类：Engineering 小类：General Engineering	Q1	9 / 275	96%
			大类：Computer Science 小类：General Computer Science	Q1	10 / 204	95%
			大类：Materials Science 小类：General Materials Science	Q1	55 / 439	87%

尹志坚——DNCNN


Preprocessing
Feature extraction
Classification
Input
3
48
F1
32
48
F2
32
48
F3
64
48
F4
64
24
F5
64
24
F6
64
24
F7
64
12
F8
384
12
F9
384
12
F10
256
12
F11
256
6
P7:2×2
P11:2×2
NC1:32@
3×3×3
NC2:32@
3×3×32
NC3:32@
3×3×32
P4:3×3
NC5:64@
3×3×64
NC6:64@
3×3×64
NC8:384@
3×3×64
NC9:384@
3×3×384
NC10:256@
3×3×384
FL12:2048
FL13:2048
Output:2
Full connection

尹志坚——DNCNN_微调超参数

Layer	Layer Type	Hyper-parameters
Input	Input	Image size: 48×48
NC1	Normalization and convolution	Filer size k1×k1: 3×3
		Filer number n1: 32
		Stride: s1×s1: 1×1
		Activation function: ReLU
NC2	Normalization and convolution	Filer size k2×k2: 3×3
		Filer number n2: 32
		Stride: s2×s2: 1×1
		Activation function: ReLU
NC3	Normalization and Convolution	Filer size k3×k3: 3×3
		Filer number n3: 64
		Stride: s3×s3: 1×1
		Activation function: ReLU
P4	Pooling	Pooling region size k4×k4: 3×3
		Stride: s4×s4: 2×2
		Pooling method: max-pooling
NC5	Normalization and convolution	Filer size k5×k5: 3×3
		Filer number n5: 64
		Stride:s5×s5: 1×1
		Activation function: ReLU
NC6	Normalization and convolution	Filer size k6×k6: 3×3
		Filer number n6: 64
		Stride: s6×s6: 1×1
		Activation function: ReLU
P7	Pooling	Pooling region size k7×k7: 2×2
		Stride: s7×s7: 2×2
		Pooling method: max-pooling
NC8	Normalization and convolution	Filer size k8×k8: 3×3
		Filer number n8: 384
		Stride: s8×s8: 1×1
		Activation function: ReLU
NC9	Normalization and convolution	Filer size k9×k9: 3×3
		Filer number n9: 384
		Stride:s9×s9: 1×1
		Activation function: ReLU
NC10	Normalization and convolution	Filer size k10×k10: 3×3
		Filer number n10 : 256
		Stride:s10×s10: 1×1
		Activation function: ReLU
P11	Pooling	Pooling region size k11×k11: 2×2
		Stride:s11×s11: 2×2
		Pooling method: max-pooling
FL12	Fully-connected	Neurons number n12: 2048
		Dropout probability p: 0.5
		Activation function: ReLU
FL13	Fully-connected	Neurons number n13: 2048
		Dropout probability p: 0.5
		Activation function: ReLU
Output	Output	Neurons number n14: 2

尹志坚——DNCNN_微调超参数

TABLE 4. Evaluation of performance with different hyper-parameters.

Network Design	Set1			Set2		
	AR(%)	DR(%)	FAR(%)	AR(%)	DR(%)	FAR(%)
k4=2	97.32	94.20	0.60	97.07	94.18	0.49
k7=k11=3	97.10	93.65	0.60	97.87	95.93	0.49
n12=n13=4096	96.39	94.20	2.16	96.75	95.05	1.83
$\varepsilon(x)=\sigma(x)$	97.78	95.10	0.84	97.33	94.91	0.61
$\varepsilon(x)=\tanh(x)$	96.67	93.47	1.20	**98.33**	**97.38**	0.85
hyper-parameters in **Table 3**	**97.83**	**95.28**	**0.48**	98.08	96.36	**0.48**

尹志坚——DNCNN_数据集

袁飞牛数据集97.83%正确率

袁飞牛博士创建的烟雾和非烟雾图像数据库：
（1）烟雾图像：黑白648；
（2）非烟雾图像：非烟雾1_27707，非烟雾2_28760；
（3）烟雾和非烟**组1**：烟雾552和无烟831，
（4）烟雾和非烟**组2**：烟雾668和无烟817，
（5）烟雾和非烟**组3**：烟雾2201和无烟8511，
（6）烟雾和非烟**组4**：烟2254和无烟8363

TABLE 1. image data sets for training, validation and testing.

Datasets	Number of smoke images	Number of non-smoke images	Total number of images	Purposes for
Set1	552	831	1383	Testing
Set2	688	817	1505	Testing
Set3	2201	8511	10712	Training
Set4	2254	8363	10617	Validation

http://staff.ustc.edu.cn/~yfn/vsd.html

张的数据集

尹志坚——DNCNN_数据扩充

两种思路：
1.比例保持不变(1:4)，原有数据集全部扩充
2.为了比例保证1:1，仅扩充有烟（正例）

扩充方式：水平翻转，垂直翻转，中心旋转

实验证明：
只有增加数据集大小，并且保证正例和负例的比例相近时，才能提高模型性能

$$\mathrm{DR} = \frac{P_p}{Q_p} \times 100\%$$

$$\mathrm{FAR} = \frac{N_p}{Q_n} \times 100\%$$

$$\mathrm{AR} = \frac{P_P + N_n}{Q_n + Q_p} \times 100\%$$

TABLE 5. Original and augmented datasets.

Datasets	Number of smoke images	Number of non-smoke images	Total number of images
Set3	2201	8511	10712
SetL	8804	34044	42848
SetB	8804	8511	17315

TABLE 6. Experiments with data augmentation.

Datasets	Set1			Set2		
	AR(%)	DR(%)	FAR(%)	AR(%)	DR(%)	FAR(%)
Set3	97.83	95.28	**0.48**	98.08	96.36	0.48
SetL	97.57	94.74	0.77	98.33	96.65	**0.24**
SetB	**98.19**	**96.37**	0.60	**98.53**	**97.52**	0.61

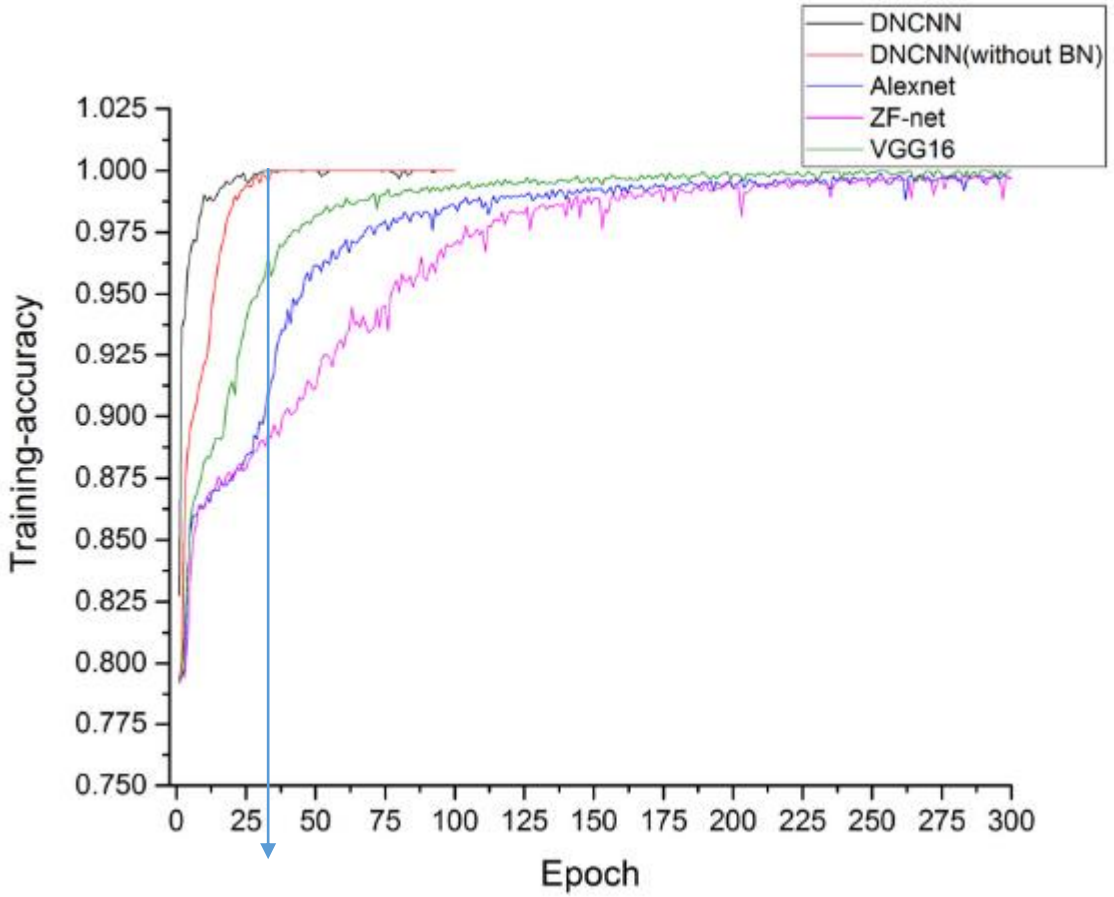
尹志坚——DNCNN_与经典网络对比

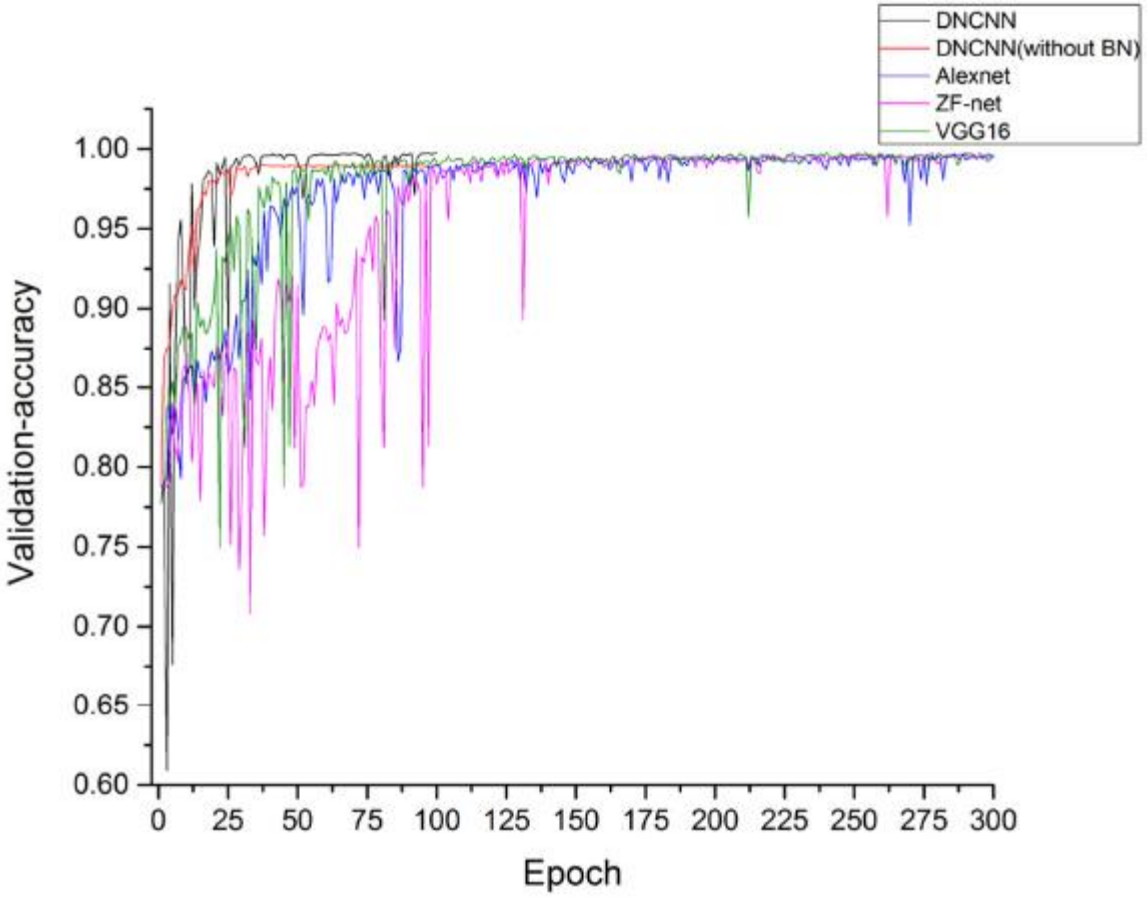
TABLE 7. Comparisons with classical deep CNNs.

Deep CNNs	Input image size	Set1			Set2			Learnable parameters
		AR(%)	DR(%)	FAR(%)	AR(%)	DR(%)	FAR(%)	
AlexNet	227	97.18	93.29	**0.24**	98.04	96.07	**0.12**	60 Million
ZF-Net	224	97.18	93.29	0.24	97.01	94.33	0.73	60M
VGG16	224	97.48	**96.19**	0.60	97.48	95.05	0.48	120M
DNCNN (without BN)	48	94.86	90.21	2.04	95.94	93.02	1.59	20M
DNCNN	48	**97.83**	95.28	0.48	**98.08**	**96.36**	0.48	20M

结论：
1.有batch normalization的，提高模型泛化能力
2.有准确率的优势，同时参数最少，有效减少算力

尹志坚——DNCNN_对比图





尹志坚——DNCNN_可视化

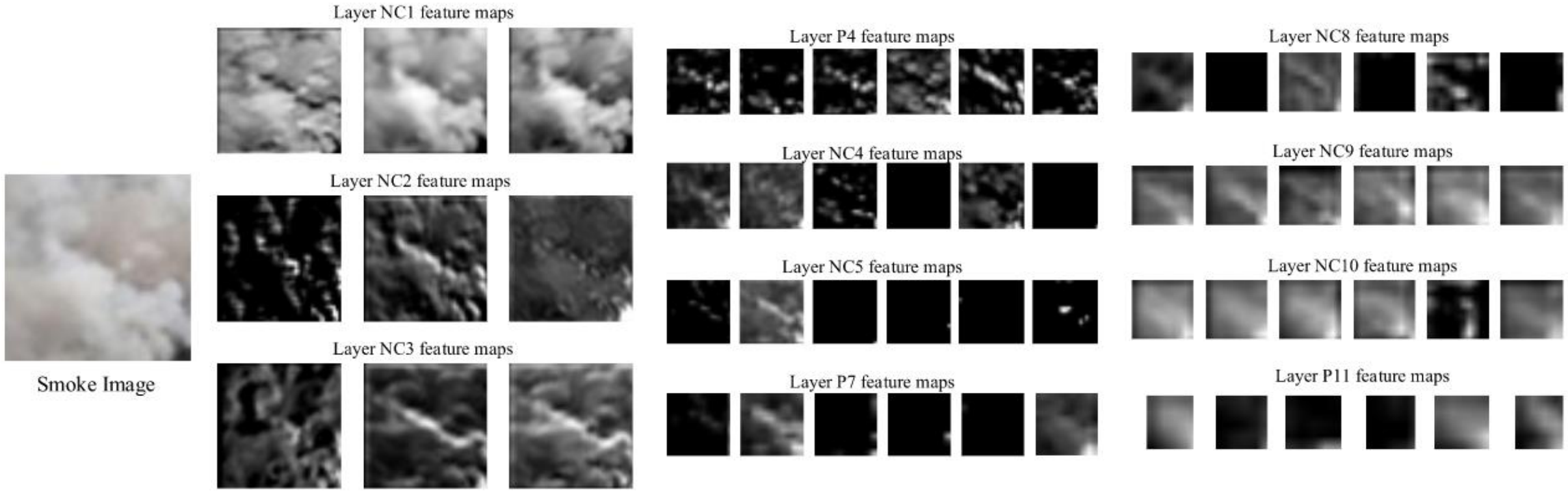


FIGURE 5. **Visualization of feature maps in convolutional and pooling layers learned by our DNCNN on SetB.**

尹志坚——DNCNN_不同FC层影响

TABLE 9. Evaluation of different classification layers.

Number of fully connected layers	Set1			Set2			Dimension of feature
	AR(%)	DR(%)	FAR(%)	AR(%)	DR(%)	FAR(%)	
3	**97.83**	**95.28**	**0.48**	**98.08**	**96.36**	**0.48**	**2048**
2	97.46	94.74	0.72	96.87	93.75	0.48	2048
1	60.08	0.00	0.00	54.28	0.00	0.00	6*6*256

A Dual-Channel convolution neural network for image smoke detection

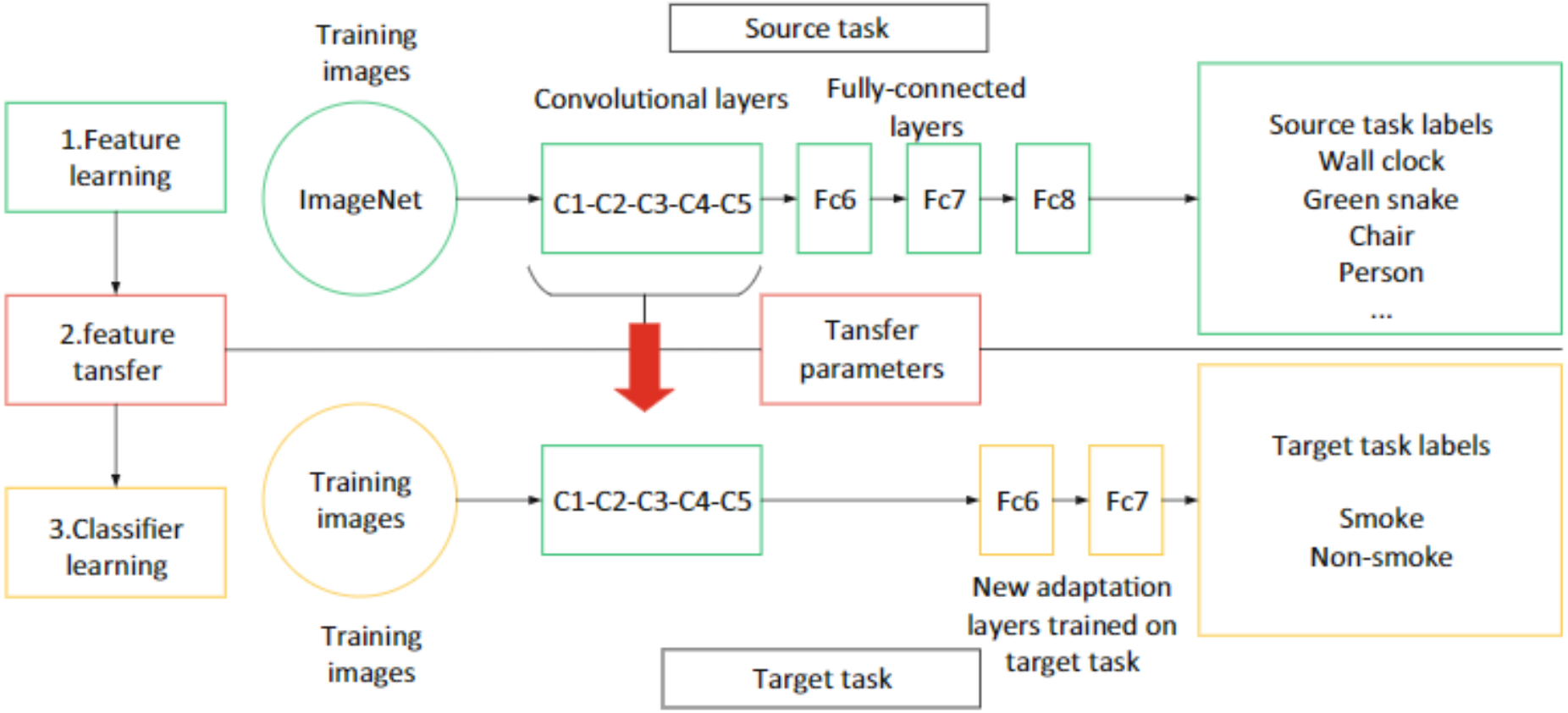
Fang Zhang [1,2] · Wen Qin [1,2] · Yanbei Liu [1,2] · Zhitao Xiao [1,2] · Jinxin Liu
Qi Wang [1,2] · Kaihua Liu [3]

Multimedia Tools and Applications

Springer

	CiteScore	SJR	SNIP	CiteScore排名			
				学科	分区	排名	百分位
CiteScore	2.33	0.335	1.038	大类：Engineering 小类：Media Technology	Q1	9 / 66	87%
				大类：Computer Science 小类：Hardware and Architecture	Q1	43 / 152	72%
				大类：Computer Science 小类：Computer Networks and Communications	Q1	80 / 274	70%
				大类：Computer Science 小类：Software	Q1	131 / 360	63%

DC-CNN 迁移学习


Source task
Training images
Convolutional layers
Fully-connected layers
1.Feature learning
ImageNet
C1-C2-C3-C4-C5
Fc6
Fc7
Fc8
Source task labels
Wall clock
Green snake
Chair
Person
...
2.feature tansfer
Tansfer parameters
Training images
C1-C2-C3-C4-C5
Fc6
Fc7
Target task labels
Smoke
Non-smoke
3.Classifier learning
New adaptation layers trained on target task
Training images
Target task

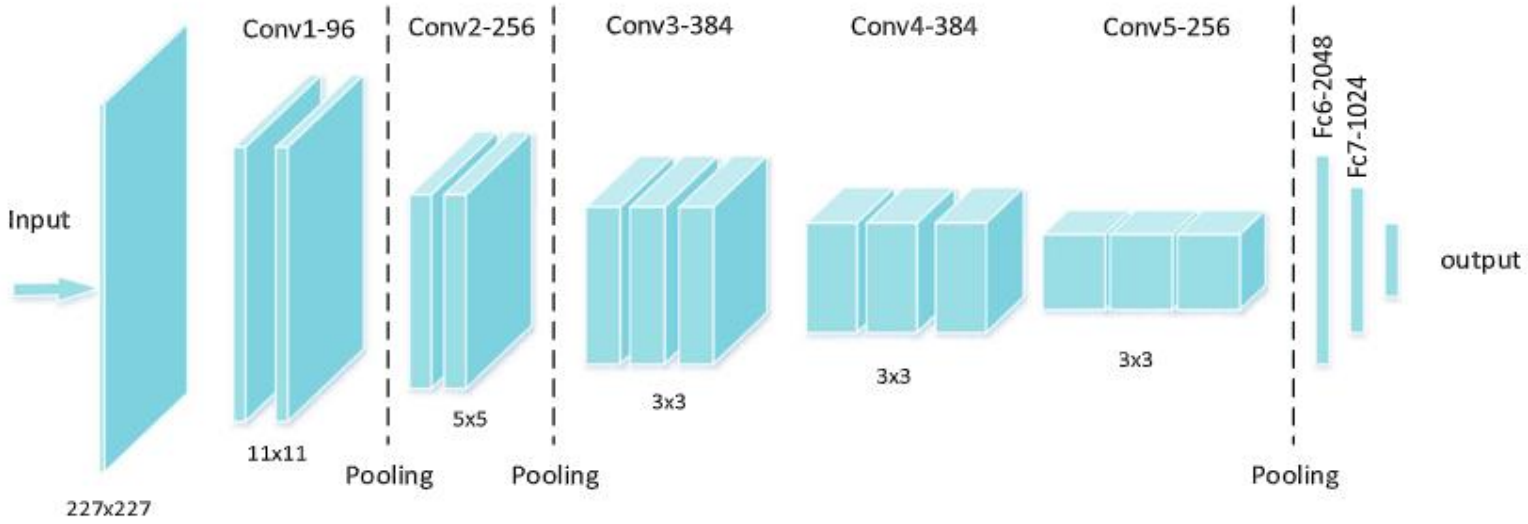


Fig. 2 The first channel: AlexNet with transfer learning

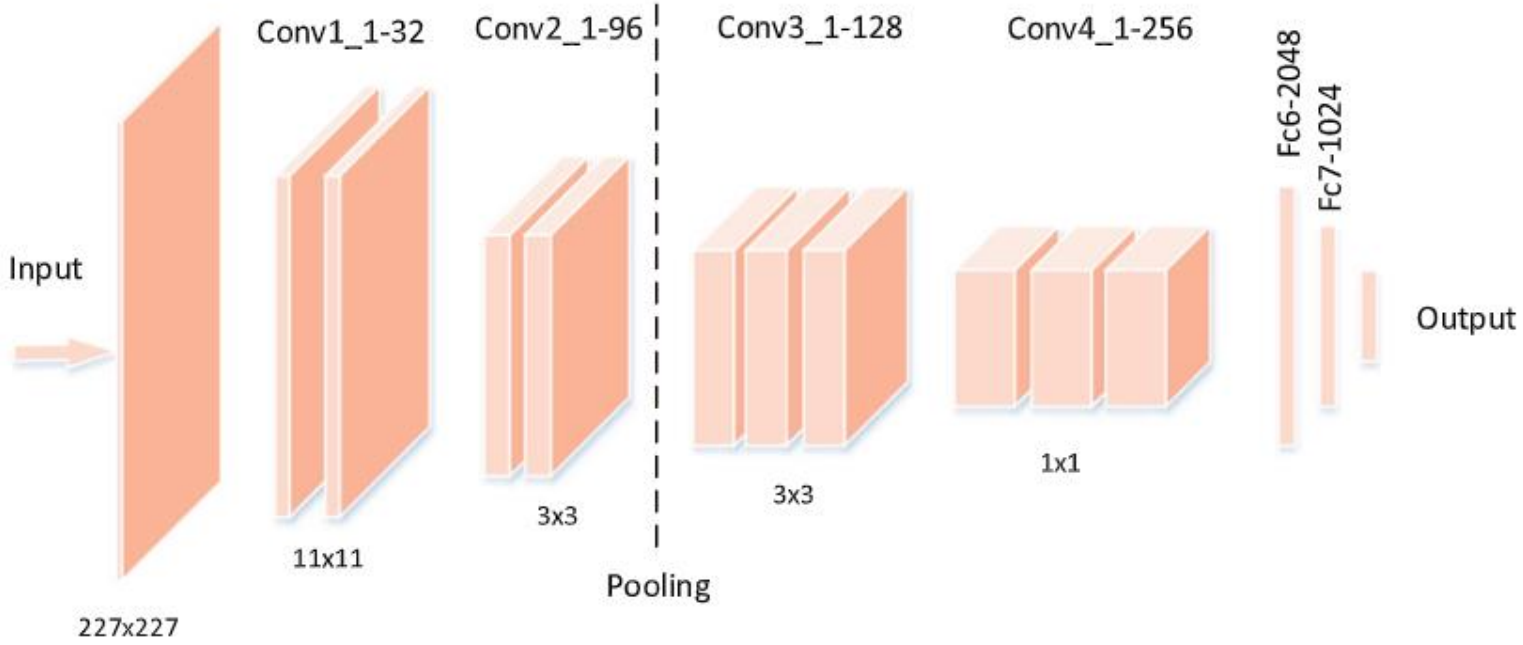
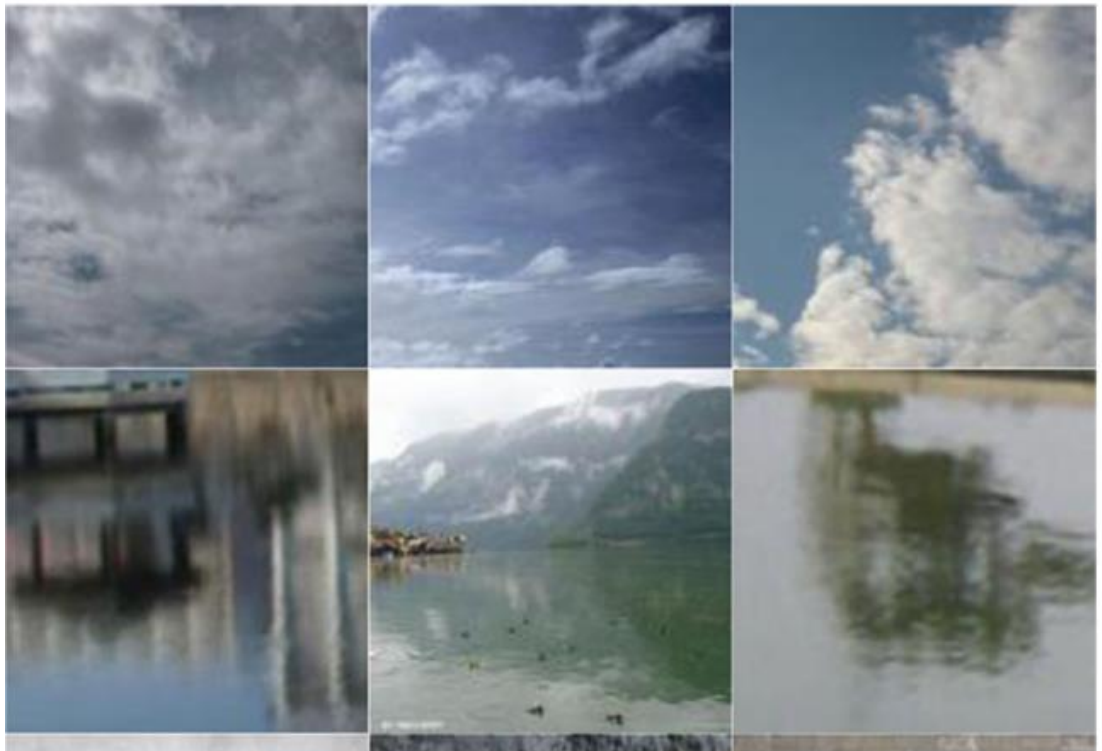


Fig. 3 The second channel of detail-extracted CNN

DC-CNN 数据集

Table 1 The image data sets for training, validation and testing

Datasets	Number of smoke images	Number of non-smoke images	Total number of images	Purposes for
Set1	7934	7934	15,868	Training
Set2	860	860	1720	Validation
Set3	1000	1000	2000	Testing



袁数据集

DC-CNN 迁移学习效果更佳

Table 4 Comparison of testing experimental results

Algorithm	Accuracy	Loss
AlexNet	0.9696	0.2284
AlexNet +transfer learning	0.9908	0.0557

DC-CNN DNCNN对比

Table 6 Comparison of testing experimental results

Deep CNNs	Input image size	Accuracy	Loss	Learned parameters
SC [4]	16×16	0.8208	–	–
LeNet	32×32	0.9444	0.2478	2 Million
AlexNet	227×227	0.9696	0.2284	216 M
VGG-16	224×224	0.9754	0.1873	258 M
DNCNN [29]	48×48	0.9826	0.1060	43 M
AlexNet+TL	227×227	0.9908	0.0557	216 M
DC-CNN	227×227	0.9933	0.0238	161 M

总结

1.数据增强——保证正例与负例1:1

2.迁移学习——有助于提高泛化能力

3.可视化——帮助理解不同层的变化

4.网络结构确定——试错法、消融实验

5.多层网络一定比单网络的识别精度高

Thank You!